



DYNAMIC PROGRAMMING-2

MARZIEH ESKANDARI
NJIT

0,1-KNAPSACK PROBLEM

$$S = \{item_1, item_2, \dots, item_n\}$$

$$w_i = Weight(item_i)$$

$$b_i = benefit(item_i)$$

$$W = Max\ Weight$$

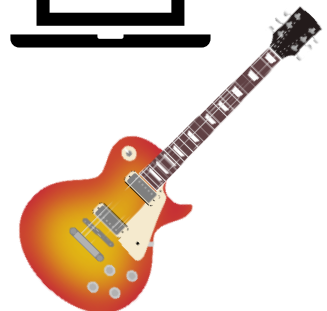
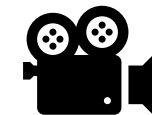
Find a subset of $A \subseteq S$, such that:

The total weight of items in A does not exceed W and the total benefit of items in A is maximized.

Item	Weight	Benefit
Camera	10	320
Laptop	20	400
Watch	8	220
Guitar	12	360



W=20 lbs



0,1-KNAPSACK PROBLEM

Problem. Find a subset of items from $S = \{item_1, item_2, \dots, item_n\}$, such that the total weight of the selected items does not exceed W and the total benefit of them is maximized.



Problem. Find the **maximum total benefit** by selecting a subset of items from $S = \{item_1, item_2, \dots, item_n\}$, such that the total weight of the selected items does not exceed W .

0,1-KNAPSACK PROBLEM: SUBPROBLEMS

Problem. Find the maximum total benefit by selecting a subset of items from $S = \{item_1, item_2, \dots, item_n\}$, such that the total weight of the selected items does not exceed W .



Subproblems: Find the maximum total benefit by selecting a subset of items from $\{item_1, item_2, \dots, item_i\}$, such that the total weight of the selected items does not exceed j and $0 \leq i \leq n$ and $0 \leq j \leq W$.

0,1-KNAPSACK PROBLEM:TABLE

Subproblems: Find the maximum total benefit by selecting a subset of items from $\{item_1, item_2, \dots, item_i\}$, such that the total weight of the selected items does not exceed j and $0 \leq i \leq n$ and $0 \leq j \leq W$.

Table: $A[i][j]$ = maximum benefit by selecting a subset of items from $\{item_1, item_2, \dots, item_i\}$, such that the total weight of the selected items does not exceed j for all i and j : $0 \leq i \leq n$ and $0 \leq j \leq W$.

0,1-KNAPSACK PROBLEM:TABLE

Subproblems: Find the maximum total benefit by selecting a subset of items from $\{item_1, item_2, \dots, item_i\}$, such that the total weight of the selected items does not exceed j and $0 \leq i \leq n$ and $0 \leq j \leq W$.

Table: $A[i][j]$ = max benefit using first i items with capacity j
 $0 \leq i \leq n$ and $0 \leq j \leq W$

0,1-KNAPSACK PROBLEM: SMALL PROBLEMS

Table: $A[i][j]$ = max benefit using first i items with capacity j
 $0 \leq i \leq n$ and $0 \leq j \leq W$

Smallest problems:

$$A[i][0] = 0: 0 \leq i \leq n,$$

$$A[0][j] = 0: 0 \leq j \leq W$$

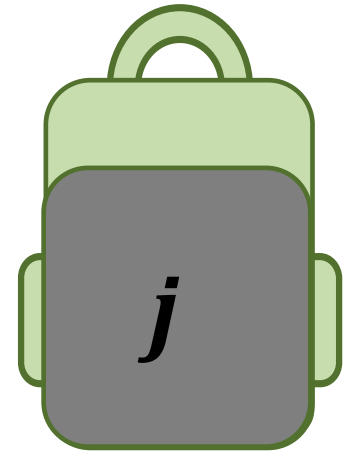
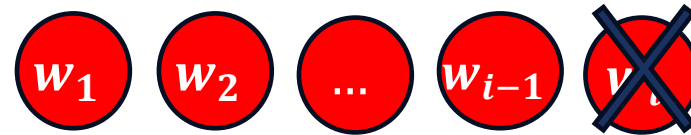
Main problem:

$$A[n][W] = ?$$

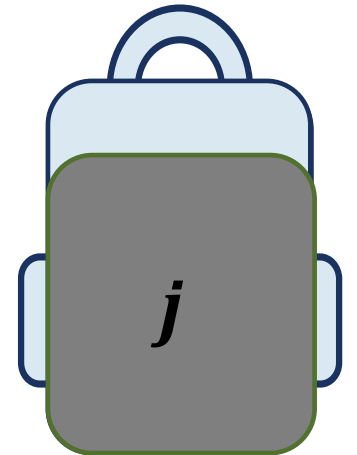
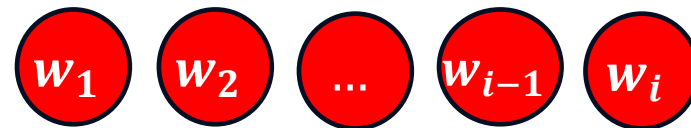
0,1-KNAPSACK PROBLEM: RELATION

1. Don't take item i:

- The best you can do is what you could do with the first $i-1$ items at full capacity j .
- So: $A[i-1][j]$



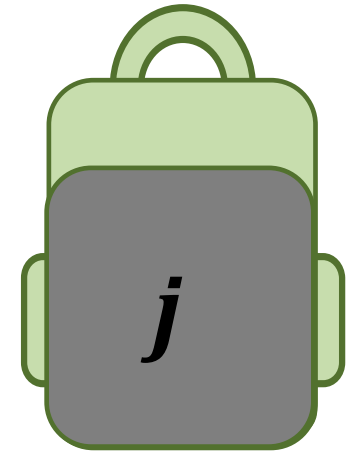
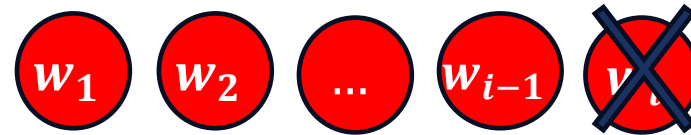
2. Take item i:



0,1-KNAPSACK PROBLEM: RELATION

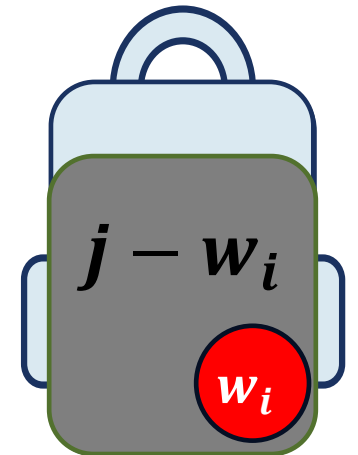
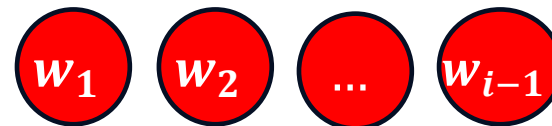
1. Don't take item i :

- The best you can do is what you could do with the first $i-1$ items at full capacity j .
- So: $A[i - 1][j]$



2. Take item i :

- You **spend** w_i units of capacity, and you're left with $j - w_i$.
- You now add the benefit of item i to the best you could do with the first $i-1$ items and this reduced capacity.
- So: $A[i - 1][j - w_i] + b_i$



0,1-KNAPSACK PROBLEM: RELATION

relation: $A[i][j] = \max\{A[i-1][j-w_i] + b_i, A[i-1][j]\}$



what if $j - w_i < 0$????

0,1-KNAPSACK PROBLEM

```
Function Knapsack01(weights, benefits, W):
```

```
    A is an nxW matrix of zeros
```

```
    for i from 1 to n:
```

```
        for j from 1 to W:
```

```
            if weights[i] <= j:
```

```
                A[i][j] ← max{A[i - 1][j], A[i - 1][j - weights[i]] + benefits[i]}
```

```
            else:
```

```
                A[i][j] ← A[i-1][j]
```

```
    return A[n][W]
```

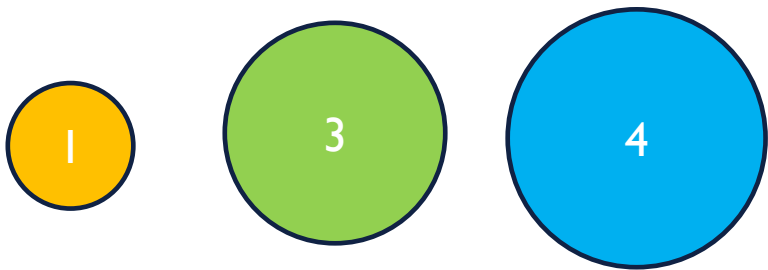


CLASS WORK



MONEY CHANGE PROBLEM

- Compute the minimum number of coins needed to change the given value into coins with denominations 1, 3, and 4.
- Input: An integer money, n .
- Output: The minimum number of coins that changes n .



$$\begin{aligned}6 &= 4 + 1 + 1 \\6 &= 3 + 3\end{aligned}$$

$$\begin{aligned}10 &= 4 + 4 + 1 + 1 \\10 &= 4 + 3 + 3\end{aligned}$$

$$n = 4k + 2$$

$$k(4) + 2(1)$$

$k + 2$ coins

$$(k-1)(4) + 2(3)$$

$k + 1$ coins

SUBPROBLEMS

- Find the minimum number of coins with denominations 1, 3 and 4 that changes i , $0 \leq i \leq n$.

TABLE

- $D[i]$ = minimum number of coins with denominations 1, 3 and 4 that changes i , $0 \leq i \leq n$.
- Smallest problem: minimum number of coins that changes 0
- $D[0]=0$
- $D[n]$ =solution

RELATION

1. Choose 
2. Choose 
3. Choose 

$D[i]$ = minimum number of coins with denominations 1, 3 and 4 that changes i

RELATION

1. Choose  $i - 1$
2. Choose  $i - 3$
3. Choose  $i - 4$

$D[i]$ = minimum number of coins with denominations 1, 3 and 4 that changes i

RELATION

1. Choose  $D[i - 1] + 1$
2. Choose  $D[i - 3] + 1$
3. Choose  $D[i - 4] + 1$

RELATION

1. Choose  $D[i - 1] + 1$
2. Choose  $D[i - 3] + 1$
3. Choose  $D[i - 4] + 1$

$$D[i] = \min\{D[i - 1] + 1, D[i - 3] + 1, D[i - 4] + 1\}$$

INDUCTIVE APPROACH

```
create array D[0..n]
```

```
D[0]=0
```

```
for i=1 to n do
```

```
    min_coins=1+D[i-1]
```

```
    if i-3>=0 then
```

```
        min_coins=min(min_coins,D[i-3]+1)
```

```
    if i-4>=0 then
```

```
        min_coins=min(min_coins,D[i-4]+1)
```

```
    D[i]=min_coins
```

```
return D[n]
```

MINIMUM STEPS TO REDUCE A NUMBER TO 1

- Given a number n , find the minimum number of steps to reduce it to 1 performing the following operations:
 1. If n is divisible by 3, divide it by 3.
 2. If n is divisible by 2, divide it by 2.
 3. Subtract 1 from n .

The goal is to minimize the number of operations. $11 \rightarrow 1$

$11 \rightarrow 10 \rightarrow 5 \rightarrow 4 \rightarrow 2 \rightarrow 1$

$11 \rightarrow 10 \rightarrow 9 \rightarrow 3 \rightarrow 1$

SUBPROBLEMS

- Given a number i , find the minimum number of steps to reduce it to 1 performing the following operations:
 1. If i is divisible by 3, divide it by 3.
 2. If i is divisible by 2, divide it by 2.
 3. Subtract 1 from i .

TABLE

- $D[i]$ = minimum number of steps to reduce i to 1 performing the given operations,
 $1 \leq i \leq n$
- Smallest Problem: $i = 1$, minimum number of steps to reduce 1 to 1 performing the given operations
- $D[1] = 0$: No steps needed to reduce 1 to 1
- Solution to the main problem is located at $D[n]$

$D[i]$ = minimum number of steps to reduce i to 1 performing the given operations

RELATION

- To calculate $D[i]$:
 1. If i is divisible by 3,
 2. If i is divisible by 2,
 3. Subtract 1 from i ,

$D[i]$ = minimum number of steps to reduce i to 1 performing the given operations

RELATION

- To calculate $D[i]$:
 1. If i is divisible by 3, divide it by 3 and get $i/3$, (1 operation), solve the problem for $i/3$.
 2. If i is divisible by 2, divide it by 2 and get $i/2$, (1 operation), solve the problem for $i/2$.
 3. Subtract 1 from i , (1 operation), solve the problem for $i - 1$.

$D[i]$ = minimum number of steps to reduce i to 1 performing the given operations

RELATION

■ To calculate $D[i]$:

1. If i is divisible by 3: $1 + D\left[\frac{i}{3}\right]$.
2. If i is divisible by 2: $1 + D\left[\frac{i}{2}\right]$.
3. $1 + D[i - 1]$.

$$D[i] = \min \begin{cases} 1 + D\left[\frac{i}{3}\right] & \text{if } i \% 3 == 0 \\ 1 + D\left[\frac{i}{2}\right] & \text{if } i \% 2 == 0 \\ 1 + D[i - 1] \end{cases}$$

$$D[i] = 1 + \min \begin{cases} D\left[\frac{i}{3}\right] & \text{if } i \% 3 == 0 \\ D\left[\frac{i}{2}\right] & \text{if } i \% 2 == 0 \\ D[i - 1] \end{cases}$$

INDUCTIVE APPROACH

```
min_steps_to_one(n):  
    create array D[1..n]  
    D[1] = 0  
    for i = 2 to n do  
        min_steps = D[i-1]  
        if i % 2 == 0 then  
            min_steps = min(min_steps, D[i/2])  
        if i % 3 == 0 then  
            min_steps = min(min_steps, D[i/3])  
        D[i] = 1 + min_steps  
    return D[n]
```

ANY QUESTION??

HAVE A NICE DAY!